INCH-POUND

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DETAIL SPECIFICATION

CLOTH, COATED, COTTON, VINYL COATED, FIRE AND MILDEW RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for a waterproof, fire and mildew resistant, vinyl coated cotton cloth.

….1.2 Classification. The cloth is available in the following types and classes as specified (see 6.2).

1.2.1 Types and classes.

Type I - Coated cloth
Class 1 - Plain weave 7.0 to 8.5 ounces (coated on both sides)
Class 2 - DELETED
Class 3 - Twill weave 14.0 to 16.0 ounces (coated on one side)

Type II - Coated duck (coated on both sides).
Class 1 - Army duck, cotton plied yarns, 15.5 to 18.5 ounces
Class 2 - DELETED
Class 3 - Army duck, cotton plied yarns, 12.0 to 15.0 ounces
Class 4 - Army duck, cotton plied yarns, 18.0 to 21.0 ounces
Class 5 - Army duck, cotton plied yarns, 22.0 to 25.0 ounces

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-8184 - Plastic Sheet, Acrylic, Modified

(Copies of this document are available online at https://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

Title 16 CFR Part 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of these documents are available online at https://ecfr.gov.)

ENVIRONMENTAL PROTECTION AGENCY

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (40 CFR Part 162) State Registration of Pesticide Products

Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) FIFRA as amended by the Food Quality Protection Act of 1996 and the Pesticide Registration Improvement Act

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(Copies of these documents are available online at https://www.epa.gov/pesticides.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter Laundering
AATCC TM22 - Test Method for Water Repellency: Spray Test:
AATCC TM30 - Test Method for Antifungal Activity, Assessment on Textile Materials: Mildew and Rot Resistance of Textiles Materials
AATCC TM70 - Test Method for Water Repellency: Tumble Jar Dynamic Absorption
AATCC TM 127 - Test Method for Water Resistance: Hydrostatic Pressure
AATCC TM169 - Test Method for Weather Resistance of Textiles: Xenon Lamp Exposure

(Copies of these documents are available online at https://aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D471 - Standard Test Method for Rubber Property-Effect of Liquids
ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics
ASTM D751 - Standard Test Method for Coated Fabrics
ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus
ASTM D2261 - Standard Test Method for Tearing Strength of Fabrics by Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Machine)
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics

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ASTM D3776/D3776M - Standard Test Methods for Mass per Unit Area (Weight) of Fabric
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6413/D6413M - Standard Test Methods for Flame Resistance of Textiles Vertical Test)
ASTM E1252 - Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis

(Copies of these documents are available online at https://www.astm.org.)

SAE INTERNATIONAL

SAE-J1966 - Lubricating Oils, Aircraft Piston Engine (Non-Dispersant Mineral Oil)

(Copies of this document are available online at https://www.sae.org/standards/content/J1966_20201.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise indicated, the finished coated cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Base cloths. The duck cloths shall be made of cotton yarn which has been carded, drawn, and spun into singles or twisted into plied yarns to meet the specified requirements. The base cloth for Type I, Class 3, shall be a greige undyed cloth and shall be napped on one side only.

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3.4.2 Weave. The weave for the Type I, Class 1 cloth shall be a plain weave. The weave for the Type I, Class 3 cloth shall be a twill weave. The weave for the Type II, all classes shall be a plain duck weave.

3.5 Color. The color of the coated colored cloth shall be as specified in the contract or procurement documents (see 6.2).

3.5.1 Visual shade matching. The color and appearance of the dyed/coated cloth shall match the standard sample (see 6.3) when tested as specified in 4.5.

3.5.2 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The cloth shall be "Free" of labile sulfur when tested as specified in 4.5.

3.5.3 Aluminized coating. When aluminized coating is required (see 6.2), it shall contain sufficient aluminized pigment to produce a bright reflecting surface equal to the standard sample.

3.6 Base cloths physical requirements. The base cloths shall conform to the requirements, listed below, in Table I, when tested as specified in 4.5.

TABLE I. Physical requirements for base cloths.

Characteristic	Requirements					
	Type I		Type II 1/			
	Class 1	Class 3	Class 1	Class 3	Class 4	Class 5
Weight, oz./yd^2						
Minimum	4.0	8.5	9.85	8.25	12.30	14.8
Maximum	4.5	-	11.0	9.25	13.50	16.30
Yarns per inch, (min.)						
Warp	-	76	-	-	-	-
Filling	-	48	-	-	-	-
Breaking strength, pounds, (min.)						
Warp	80	130	160	125	210	235
Filling	80	75	110	120	130	175
Tearing strength, pounds, (min.)						
Warp	-	-	-	4.0	-	-
Filling	-	-	-	3.0	-	-
Air permeability, ft^3/min/ft^2, (max)	-	-	5.0	4.0	5.0	-

1/ The warp yarns shall be plied and the filling yarns shall be singled or plied.

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3.7 Mildew resistant (not applicable to Type I, Class 3 cloth). The cotton duck cloths shall be mildew resistant treated by applying treatment to either base cloth or to the coating compound. The cloth shall be treated with an inorganic, metalized, mildew inhibitor. The treated cloth shall retain a 75% breaking strength after 12-weeks soil burial initially and after leaching when tested as specified in 4.5

3.8 Coating compound.

3.8.1 All types and classes (except Type I, Class 3). The coating compound shall be formulated using a virgin vinyl chloride polymer or a virgin vinyl chloride acetate copolymer to meet mildew resistance requirements for Type I, Class1 and mildew and flame resistance for Type II all classes as specified. The coating shall not contain any mercury compounds.

3.8.1.1 Coating application. The coating compound shall be applied to both sides of the cloth at a rate to meet overall weight requirements specified in Table I. One side of the cloth shall be just sufficiently coated to meet the resistance to water absorption requirement and the balance of the coating shall be applied to the other side. Calender coating with a preformed film shall not be permitted. Unless otherwise specified, the heavily coated side shall be the face.

3.8.2 Type I, Class 3. The coating for the Type I, Class 3 shall be a vinyl resin suitably compounded and pigmented to meet the requirements of this document.

3.8.2.1 Coating application. The coating for the Type I, Class 3 shall be applied only to the unnapped side of the cloth.

3.8.3 Physical requirements. The coated cloths shall conform to the requirements, listed below, in Table II, when tested as specified in 4.5.

TABLE II. Physical requirements for coated cloths.

Characteristic	Requirements					
	Type I		Type II			
	Class 1	Class 3	Class 1	Class 3	Class 4	Class 5
Weight, oz./yd^2						
Minimum	7.0	14.0	15.5	12.0	18.0	22.0
Maximum	8.5	16.0	18.5	15.0	21.0	25.0
Breaking strength, pounds, (min.)						
Initial						
Warp	80	130	160	125	210	235
Filling	80	75	110	120	130	175
After accelerated weathering						
Warp	72	117	112	88	147	165
Filling	72	66	77	84	91	123

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TABLE II. Physical requirements for coated cloths. - Continued

Characteristic	Requirements					
	Type I		Type II			
	Class 1	Class 3	Class 1	Class 3	Class 4	Class 5
Tearing strength, (min.)						
Elmendorf, grams						
Warp	-	-	2400	-	-	-
Filling	-	-	1900	-	-	-
Tongue method, pounds						
Warp	2.0	6.0	-	3.0	5.0	5.0
Filling	2.0	6.0	-	3.0	5.0	5.0
Stiffness, cm, (max)						
Warp only						
At 70° F	-	-	13.0	-	-	-
At -40° F	-	-	16.0	-	-	-
Adhesion of coating, pounds per 2-inch width (min.)	6.0	6.0	6.0	6.0	8.0	8.0
Blocking, scale rating, (min.)	2	3	3	3	3	3
Resistance to water absorption, percent, (max)	-	-	5.0	5.0	5.0	5.0
Crocking, (min.)						
Wet	-	-	3.5	-	-	-
Dry	-	-	3.5	-	-	-
Hydrostatic resistance, mL						
Initial	No leakage	No leakage	No Leakage	No leakage	No Leakage	No Leakage
After abrasion (max)	5	5	5	5	5	5

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TABLE II. Physical requirements for coated fabric - Continued.

Characteristic	Requirements					
	Type I		Type II			
	Class 1	Class 3	Class 1	Class 3	Class 4	Class 5
Flame resistance, (Type II only), (max) 1/						
After flame, seconds						
Warp	-	-	2.0	2.0	2.0	2.0
Filling	-	-	2.0	2.0	2.0	2.0
Char length, inches						
Warp	-	-	3.5	3.5	3.5	3.5
Filling	-	-	3.5	3.5	3.5	3.5
Resistance to cold crack	No cracking, flaking, or separation of coating	No cracking, flaking, or separation of coating	No cracking, flaking, or separation of coating	No cracking, flaking, or separation of coating	No cracking, flaking or separation of coating	No cracking, flaking, or separation of coating

1/ Testing shall be conducted initially and after being exposed to150 hours of accelerated weathering.

3.8.3.1 Aromatic hydrocarbon resistance (all Types and Classes). After immersion in aromatic fuel, the coated cloths shall show no cracking, flaking, or separation of the coating from the cloth and when subjected to a static head of 20 inches of water for 10 minutes, the seepage of water shall not exceed 5 mL through the 4 1/2-inches diameter test area, when tested as specified in 4.5.

3.8.3.2 Oil resistance (Type II cloths only). The cloth shall show no evidence of seepage of oil through the material when tested as specified in 4.5.

3.8.3.3 Resistance to accelerated weathering (all types and classes). After weatherometer exposure, the coated cloth shall show no cracking or crazing when folded sharply on itself. The color change of coated cloth after accelerated weathering shall be a rating of no less than “3.0” when tested as specified in 4.5.

3.8.3.4 Crazing and discoloration (Type I, Class3 cloth only). The cloth shall not contain any materials or impurities which will cause crazing or discoloration of transparent molded plastic sheet when tested as specified in 4.5.

3.9 Face identification. The face side of the cloth shall be identified by stamping that side with the word “FACE” at each end of the roll.

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3.10 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width exclusive of the selvage. It is permissible to trim the edges of the coated cloth provided the minimum width of the finished coated cloth is maintained.

3.11 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 60 yards. The maximum pieces per roll shall be three and no single piece shall be less than 20 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.12 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with Title 16 CFR Part 303, Rules and Regulations Under the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3.2 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract or purchase order, except where otherwise indicated (see 6.2).

4.4 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Type II of MIL-STD-3064 for Type I, Class 1 and Type II cloths and it shall be examined for defects in accordance with Type I and Type II of MIL-STD-3064 for Type I, Class 3 cloth.

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4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished coated cloth for all physical and chemical tests. The lot shall be unacceptable if one test fails to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. End item tests.

Component	Characteristic	Requirement reference	Test method
Basic cloth	Weight	Table I	ASTM D3776/D3776M, Opt C
	Yarns per inch	Table I	ASTM D3775
	Breaking strength	Table I	ASTM D5034
	Tearing strength	Table I	ASTM D1424
	Air permeability	Table I	ASTM D737
Coated cloth	Color	See 3.5.1	See 4.6.1
	Sulfur	See 3.5.2	See 4.6.2
	Mildew inhibitor Initial After leaching	See 3. 7	 AATCC TM30, Test I Appendix 1A of AATCC TM30 and AATCC TM30,Test I
	Coating compound	See 3.8	ASTM E1252 (See 6.5)
	Weight	Table II	ASTM D3776/D3776M

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TABLE II. End item tests. - Continued

Component	Characteristic	Requirement reference	Test method
Coated cloth (continued)	Breaking strength, Initial After accelerated weathering	Table II	ASTM D5034 AATCC TM169, Opt 1 & ASTM D5034 1/
	Tearing strength Elmendorf Tongue method	Table II	ASTM D1424 ASTM D2261
	Stiffness At 70° F At -40° F	Table II	See 4.6.3
	Adhesion of coating	Table II	ASTM D751
	Blocking	Table II	ASTM D751
	Resistance to water absorption	Table II	AATCC TM22 & See 4.6.4
	Crocking, Wet Dry	Table II	AATCC TM8 2/
	Hydrostatic resistance Initial After abrasion	Table II	AATCC TM127 3/ See 4.6.5 & AATCC TM 127
	Flame resistance (Type II only) Initial After accelerated weathering	Table II	ASTM D6413 AATCC TM169, Opt 1 & ASTM D6413 1/
	Resistance to cold crack	Table II	See 4.6.6 ASTM D 751
	Aromatic hydrocarbon resistance	See 3.8.3.1	See 4.6.7 and AATCC TM127
	Oil resistance (Type II only)	See 3.8.3.2	See 4.6.8
	Resistance to accelerated weathering	See 3.8.3.3	AATCC TM169, Opt 1, 4/
	Crazing and discoloration (Type I, Class 3 only)	See 3.8.3.4	See 4.6.9

1/ The exposure time shall be 150 hrs.
2/ Rated using the AATCC EP8.

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3/ The lightly coated or uncoated side will be facing the water. The water height shall be raised to 20 inches and maintained at that level for 10 minutes and then examined for leaks.
4/ Color change shall be rated using AATCC EP1.

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloths shall match the standard sample when viewed using the AATCC EP9, Option C (see 6.4), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent illuminant A at 2856K (± 200).

4.6.2 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail

4.6.3 Stiffness test. Stiffness testing shall be performed in accordance with the following procedure, except that the apparatus and test specimens shall be conditioned for a minimum of four (4) hours before testing and the test shall be performed in a still atmosphere at specified temperature.

Procedure

Five (5) specimens from the warp direction only shall be cut measuring 1 1/4-inches by 6 to 12-inches with the longer dimension parallel to the direction of the warp yarns being tested. Testing shall be performed at 70°F, and -40°F using a Clark Paper Softness-Stiffness tester.

The measure of directional flex-stiffness shall be the length of the specimen remaining above the nip of the rolls which, when the test instrument is rotated clockwise and counter-clockwise through

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90 (± 2) degrees, just falls to the left and right of the line perpendicular to the nip of the rolls. It is this length of specimen which yields the flexibility criteria being measured. The specimen shall be considered to have fallen to the right or left when the specimen, during the course of clockwise and counter-clockwise rotation, passes across the line perpendicular to the nip of the rolls as indicated by the pointer fastened to the rotating framework.

The length of the specimen shall be determined as follows:

Initial adjustment of test instrument.

1. The apparatus shall be placed in the test position and leveled with the axis of the clamping surface (nip of the rolls) horizontal and the pointer, fastened to the rotating framework, in the vertical position.

2. One end of the specimen shall be inserted between the rolls with the lengthwise edges of the specimen perpendicular to the nip of the rolls.

3. The specimen shall be positioned and shall be of sufficient length to allow it to bend to the left of the line perpendicular to the nip of the rolls, which when the apparatus is rotated, with continuous uniform motion in the clockwise direction it will fall to the right of the perpendicular line.

4. At the point the specimen falls to the right, the instrument shall be stopped and the adjustable circular scale moved to make the zero position on the scale coincide with the pointer fastened to the rotating framework. The instrument shall then be rotated with continuous uniform motion (± 2) degrees from this point in the counter-clockwise direction. The specimen should not fall to the left of the perpendicular line at this point.

5. If the specimen falls to the left then the instrument shall be rotated in the clockwise direction until the pointer is vertical. The specimen shall be shortened and reset with a bend to the left and the above procedure in 4. shall be repeated.

Determination of flex-stiffness.

1. After initial adjustment the instrument is rotated until the pointer is vertical. The length of the specimen is shortened by a small increment, set with a bend to the left and the instrument is rotated with continuous uniform motion clockwise until the specimen falls to the right to the perpendicular line. The instrument is stopped and the zero position of the adjustable scale is made to coincide with the pointer. The instrument is rotated, with continuous uniform motion, in the counter-clockwise direction 90 (± 2) degrees when the instrument is immediately reversed in the clockwise direction and rotated, with continuous uniform motion, to return the pointer to zero. The specimen shall have fallen immediately to the left of perpendicular at 90 (± 2) degrees and shall fall immediately to the right of the perpendicular as the pointer reaches zero.

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2. If the specimen does not fall to the left when it is rotated the required 90 (± 2) degrees in the counter-clockwise direction the pointer is rotated to the vertical position, the specimen is shortened another small increment and the procedure in 1. shall be repeated.

3. When the specimen has reached the length such that it falls to the left and right of the line perpendicular to the nip of the rolls the instrument is stopped with the pointer in a vertical position.

4. Without tension the specimen above the rolls shall be extended and the length measured from the nip of the rolls to the end of the extended specimen, to the nearest mm.

5. The length of the extended specimen is recorded as the directional flex-stiffness of the specimen.

Report of results.

The directional flex-stiffness of the sample unit tested shall be reported as the average of the specimens tested in the warp direction. Individual values used to calculate the average shall also be reported.

4.6.4 Water absorption test. The water absorption test shall be conducted as specified in AATCC TM22 with the following exceptions:

1. The distance from the end of the funnel to the specimen shall be 24 inches and the amount of water poured shall be 500 mL.

2. Prior to exposing the three (3) specimens to the water, the specimens shall be rolled together and weighed to two (2) decimal points. This weight is designated as the original "O" weight.

3. Once the specimens have been exposed to the water spray, they shall be removed from the embroidery hoop quickly but carefully and each placed between two (2) sheets of blotting paper and placed thru two metal rollers as specified in AATCC TM70 without application of any pressure/weight.

4. The three (3) specimens shall then be rolled together and weighed to two (2) decimal points. This weight is designed as the final "F" weight.

5. Water absorption shall be calculated as follows:

$$\text{Water absorption, percent} = \frac{F - O}{O} \times 100$$

4.6.5 Abrasion test. Three (3) specimens, 10 by 10 inches shall be used for the test. The specimen shall be abraded by means of a 2-inch square piece of grade 1/C garnet paper, which shall be uniformly loaded with an 8 (± 0.1) ounce weight. The specimen shall be abraded on the heavily coated side by moving the weighted Garnet paper filling-wise five (5) times in each direction. The abraded sample shall then be positioned with the abraded area facing the water of the hydrostatic tester. Testing shall be conducted in accordance with

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AATCC TM127. The water height shall be raised to 20 inches and held for 10 minutes. If more than five (5) mL passes through the abraded area. The sample shall be reported as failing.

4.6.6 Cold crack test (Type I, Classes 1 and 3; Type II, Classes 3, 4 and 5). A 1 by 4-inch specimen with the long direction warpwise and a 1 by 4-inch specimen with the long dimension filling wise shall be exposed for four (4) hours at a temperature of minus 40 (± 2)°F, and then bent sharply, face out, over a 1/8-inch diameter mandrel so that the back of the specimen touches within 1/8 inch distance directly behind the rod. The specimen shall be examined visually for evidence of cracking, flaking or separating of the coating from the cloth.

4.6.7 Aromatic hydrocarbon resistance test. A sample 8 by 8-inches shall be immersed for five (5) minutes in 40 percent aromatic fuel. The composition of the fuel shall conform to ASTM D471, Reference fuel D. The sample shall be allowed to dry at room temperature for 2 hours (± 5) minutes, and then shall be creased at room temperature as specified in ASTM D 751. The sample shall be examined visually. Any presence of cracking, flaking or separation of coating from the cloth shall be cause for rejection. The specimen shall then be tested for hydrostatic resistance in accordance with AATCC TM 127 with the creased area positioned downward to form the centerline of the 4 1/2-inches diameter test area. The water height shall be raised to 24 inches and maintained at that level for 10 minutes.

4.6.8 Oil resistance test. Form a pocket with the cloth to which a lubricating oil, conforming to military grade 1100 of SAE-J1966 shall be added to a depth of 1/2 inch and held for 1-hour. Examine the outside for evidence of seepage of oil.

4.6.9 Test for crazing and discoloring of plastics. Test apparatus consisting of a suitable type of compression test device having top and bottom steel plates spaced 6-inches apart vertically shall be used. Four (4) specimens of plexiglass, specified in MIL-PRF-8184, 6 1/8-inches long and 1 1/2-inches wide shall be cut from approximately 5/32-inches thick stock. Three (3) specimens of the finished coated cloth shall be cut the same length and width and placed on one (1) side of each of three (3) plexiglass specimens and secured at each end with the pressure sensitive tape. The test specimen shall then be placed vertically between the ridged steel plates so the tension stresses are created on the surface of the plexiglass next to the napped back side of the cloth. A control specimen, consisting of one (1) piece of plexiglass without the cloth attached shall be placed in a similar apparatus at least 36 inches from the other three (3) specimens. Test specimens shall be left in this manner for a period of 72 hours in an atmosphere of 70 (± 2)°F and relative humidity of 65 (± 2) percent. After 72 hours, specimens shall be compared visually with the control specimen and shall not show greater crazing or discoloration than the control specimen.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military

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Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

Type I cloths are intended for use in the manufacturing of the following:

Class 1 – Airplane wing covers, engine covers and shelters.

Class 3 – Covers for molded plastic parts such as aircraft turrets, canopies and equipage items.

Type II cloths are intended for use in the manufacturing of the following:

Class 1 - Insulated frame type tents.

Class 3, 4 and 5 - Recognition panels; covers where water, oil and gasoline-proof material is required; carrying cases for life rafts; life raft equipment and similar items.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. The specific issue of individual documents referenced (see 2.1).
d. Color of cloth required (see 3.5).
e. When aluminized coating is required (see 3.5.3).
f. Width of cloth required (see 3.10).
g. Length required if other than specified (see 3.11).
h. Inspection conditions (see 4.2).
i. Conformance inspection (see 4.3.2).
j. Packaging (see 5.1).

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.5 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

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6.6 Subject term (key word) listing.

Aluminized
Army duck
Coating compound
Crazing

6.7 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing Activity:
DLA -CT

Review activities:
Army –AR, CR, MD, MI
Navy – MC, OS

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

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